```
0001 PROGRAM SIMULATION
0002 VAR
0003     LVLSCALE: LIN_TRAFO;
0004     TMPSCALE: LIN_TRAFO;
0005     SimTimer: TON;
0006     SelfRep: BOOL;
0007 END_VAR
```

```
0001
0002 (*PID for LVL and TMP*)
0003 LVLSCALE(IN:= LVL_PID, IN_MIN:= 0, IN_MAX:= 16383, OUT_MIN:= 0, OUT_MAX:= 100);
0004 TMPSCALE(IN:= TMP_PID, IN_MIN:= 0, IN_MAX:= 16383, OUT_MIN:= 0, OUT_MAX:= 1000);
0005
0006
0007 (*Note: TMP is set at room temp 70F and assumes water coming is at 70 degrees*)
0008 SimTimer(IN:= SelfRep, PT:= T#500ms);
0009 SelfRep:= TRUE;
0010 SimTimer;
0011
0012 (*Sim calculation: not based on engineering calculations*)
0013 IF Pump_Output THEN
0014     FILLRATE:= 70;
0015 ELSE
0016     FILLRATE:= 0;
0017 END_IF
0018
0019 IF Valve_Output THEN
0020     DRAINRATE:= 50;
0021 ELSE
0022     DRAINRATE:= 0;
0023 END_IF
0024
0025 IF Heater_Output THEN
0026     HEATUP:= 70;
0027 ELSE
0028     HEATUP:= 0;
0029 END_IF
0030
0031 (*70.5 is used to prevent TMP from dropping below 70 due to simulation controls calculations*)
0032 IF TMP > 70.5 THEN
0033     IF PUMP_OUTPUT THEN
0034         DROPHEAT:= 50;
0035     ELSE
0036         DROPHEAT:=  10;
0037     END_IF
0038 ELSE
0039     DROPHEAT:= 0;
0040 END_IF
0041
0042 IF SimTimer.Q THEN
0043     LVL_PID:= LVL_PID + FILLRATE - DRAINRATE;
0044     TMP_PID:= TMP_PID + HEATUP - DROPHEAT;
0045
0046     LVLSCALE.IN:= LVL_PID;
0047     LVLSCALE;
0048     LVL:= LVLSCALE.OUT;
0049     TMPSCALE.IN:= TMP_PID;
0050     TMPSCALE;
0051     TMP:= TMPSCALE.OUT;
0052     SelfRep:= FALSE;
0053     SimTimer;
0054 END_IF
0055
```

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